

E-LIBRARY MANAGEMENT SYSTEM

Project Understanding &
Implementation Document

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Academic Year: 2025-26

Internship Duration: 8 Weeks

Date: 29 May 2026

Table of Contents

1. Introduction
2. Problem Statement
3. Objective of the Project
4. Understanding of Project
5. Existing vs Proposed System
6. Features of the System
7. System Architecture
8. System Workflow
9. Technologies Used
10. Implementation Approach
11. Database Overview
12. Future Enhancements
13. Conclusion

1. Introduction

The E-Library Management System is a full-stack web-based application developed to provide an efficient and user-friendly digital platform for accessing academic and learning resources in the form of PDF documents. The system is designed to improve the traditional library process by enabling users to access study materials online in a more organized, convenient, and interactive manner.

The application allows users to register and securely log in using valid credentials. After successful authentication, users can browse available resources, search for required PDF documents, and read them through the digital platform. The system is designed to maintain user activity records, including previously accessed PDFs and reading progress.

A key feature of the proposed system is reading progress tracking and resume functionality. The application stores the last visited page of a PDF so that users can continue reading from the same point during future sessions. This eliminates the need to repeatedly search for documents or manually identify previously completed sections, thereby enhancing the overall user experience.

The project is implemented using a full-stack web development approach, where the frontend manages user interaction and interface design, the backend handles authentication, application logic, and request processing, and the database stores user information, PDF details, activity history, and reading progress records.

2. Problem Statement

Traditional library systems mainly depend on manual management and physical resource handling. Users must visit the library physically to access books, maintain borrowing records, and search for required resources manually. This process becomes time-consuming and difficult to manage efficiently.

Even in some digital library systems, users face limitations such as lack of proper activity tracking, absence of resume-reading functionality, and difficulty in organizing digital PDF resources effectively.

There is a need for a user-friendly and efficient digital library platform that allows secure access to PDF-based learning resources, quick searching, activity tracking, and reading progress management.

The proposed E-Library Management System is designed to overcome these limitations by providing an organized, accessible, and full-stack web-based digital learning environment.

Challenges in Existing Systems:

- Manual library management
- Physical visit required
- Time-consuming resource search
- No reading progress tracking
- No resume-reading feature
- Difficult resource organization

3. Objective of the Project

- Develop a digital library platform
- Provide secure user authentication
- Enable PDF search and access
- Track user reading progress
- Support resume-reading functionality
- Demonstrate full-stack web development concepts

4. Understanding of the Project

The E-Library Management System is a full-stack web application designed to provide users with easy access to digital learning resources in the form of PDF documents. The system allows users to securely access, search, read, and manage digital resources through a web-based platform.

Main Functions of the System:

- User registration and secure login
- Search and browse PDF resources
- Read PDFs within the application
- Track reading progress and user activity
- Resume reading from the last visited page
- Manage digital resources efficiently

The system integrates frontend, backend, and database technologies to provide a secure, organized, and user-friendly digital library experience.

5. Existing System vs Proposed System

Existing System	Proposed System
Manual library process	Digital web-based system
Physical book handling	PDF-based resource access
No reading progress tracking	Reading progress tracking
No resume reading feature	Resume reading Feature
Time-consuming searching	Quick search functionality
Limited accessibility	Online accessibility anytime

6. Features of the System

- **User Registration & Authentication**

The system allows users to securely register and log in using valid credentials.

- **PDF Search**

Users can search for required PDF documents using document names or keywords.

- **Digital Resource Access**

The application provides online access to PDF-based academic and learning resources.

- **Reading Progress Tracking**

The system stores the last visited page of a PDF document during reading sessions.

- **Resume Reading Functionality**

Users can continue reading from the previously saved page without restarting from the beginning.

- **Activity Tracking**

The system maintains records of previously accessed PDF documents and reading history.

- **Resource Management**

PDF documents are managed in a structured and organized digital environment.

7. System Architecture

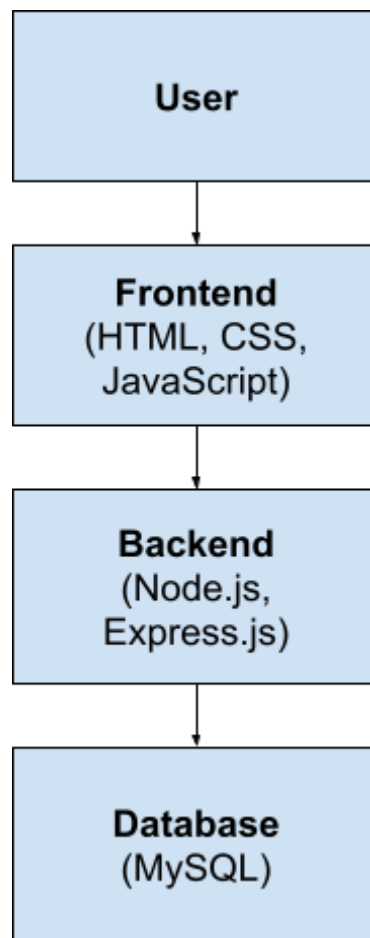
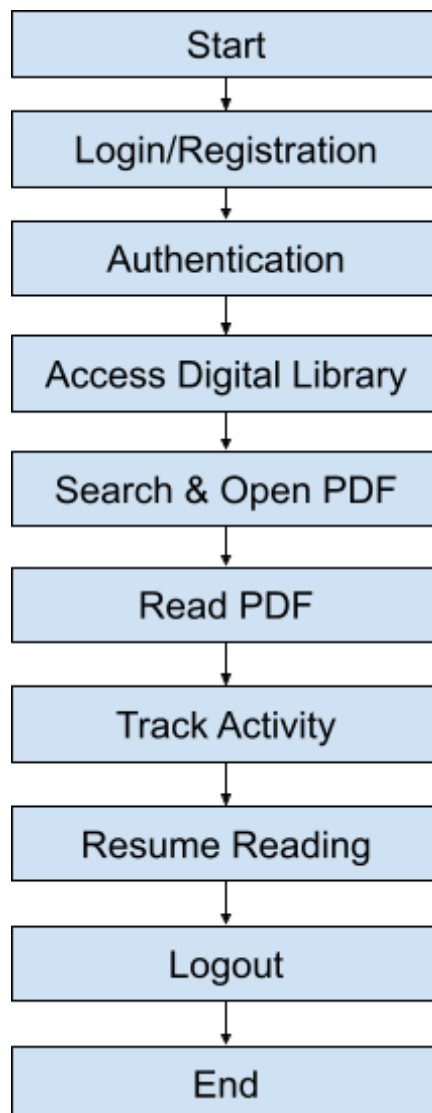


Figure 1: Three-Tier Architecture of E-Library Management System

Working of the System Architecture

- **Frontend (HTML, CSS, JavaScript):** Provides the user interface for registration, login, PDF search, and reading.
- **Backend (Node.js, Express.js):** Handles authentication, request processing, and communication with the database.
- **Database:** Stores user information, PDF details, reading history, and activity records.
- **System Integration:** All components work together to provide a smooth and secure digital library experience.

8. System Workflow



9. Technologies Used

Technology	Purpose
HTML, CSS, JavaScript	Frontend development and user interface
Node.js	Backend server-side development
Express.js	API and request handling
MySQL	Database management
VS Code	Code Editor and Development environment

10. Implementation Approach

1. User Registration and Login

The user first registers and logs into the system using valid credentials. Authentication is performed through the backend and verified using database records.

2. User Authentication

The backend validates user information and provides secure access to the digital library platform.

3. Accessing the Digital Library

After successful login, users can access the homepage containing available PDF resources and categories.

4. Searching and Opening PDFs

Users can search for required PDF documents using keywords or document names and open them within the application.

5. Reading and Tracking Activity

The system tracks user activity such as accessed PDFs and reading progress during each session.

6. Resume Reading Functionality

Previously saved reading progress is retrieved from the database so that users can continue reading from the last visited page.

7. Database Management

The database stores user details, PDF information, reading history, and activity tracking records for efficient system management.

11. Database Overview

The database stores user information, PDF details, reading history, and activity tracking records required for system operation.

Database Table	Purpose
Users Table	Stores user registration and login details
PDF Table	Stores PDF document Information
Activity Table	Maintain user activity history
Reading progress table	Stores last visited reading page
Authentication Records	Maintains secure access validation

12. Future Enhancements

- Implementation of personalized PDF recommendations based on user activity.
- Addition of bookmark and note-taking functionality for study purposes.
- Integration of notification and reminder systems for reading schedules.
- Development of an admin dashboard for managing users and PDF resources.
- Integration of cloud storage for scalable digital resource management.
- Development of a mobile-friendly version of the application.

13. Conclusion

The E-Library Management System is designed to provide a digital platform for accessing and managing PDF-based academic resources efficiently. The project demonstrates the practical implementation of full-stack web development concepts such as frontend design, backend processing, authentication, database management, and user activity tracking.

The system simplifies the traditional library process by enabling users to securely access digital resources, search for required documents, track reading progress, and resume reading from previously visited pages. The project also provides a structured and user-friendly approach for managing digital learning materials in an online environment.

Overall, the proposed system represents an efficient and scalable solution for modern digital library management and demonstrates the practical application of real-world web development technologies.